Sensor and Systems
Inertial Measurement Units

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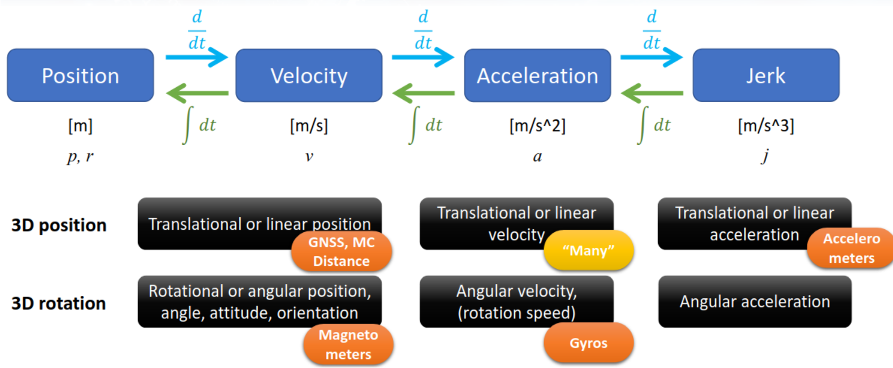
Electronic Systems

8th semester

Overview

- Inertial Measurement Units
  - Applications
  - Device configuration
- Measurement principles
  - Linear acceleration
  - Angular velocity
- Measurement model
  - Error sources
  - Model-based filtering

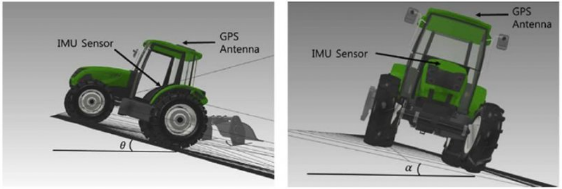
Inertial Sensors



Inertial Measurement Units

Position error by attitude angle accuracy

Terrain angle in degrees	0.1	0.5	1.0
GNSS antenna position correction in cm	0.45	1.2	4.4



- IMUs combine accelerometers, gyroscopes, and magnetometers.
- They provide high-frequency data (100 Hz to 1000 Hz).
- Faster than cameras, LiDARs, or depth sensors.
- Primary tool for short-term state estimation in robotics.
- Essential for process updates in Kalman filters.

Image credit: McLennan

Inertial Measurement Unit Applications

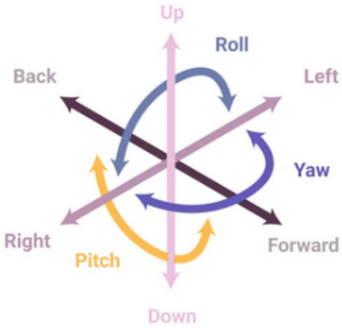
Wii

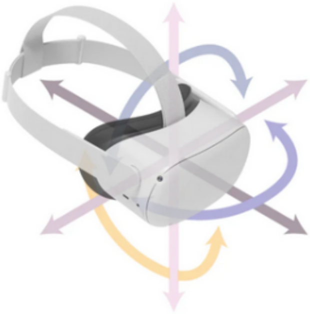
Oculus

Hisar

- Autonomous aerial drones and underwater platforms.
- Self-navigating ground vehicles and rockets.
- Virtual and Augmented Reality (e.g., Oculus headsets).
- Game consoles like the Nintendo Wii.
- Inertial navigation for aircraft and missiles.

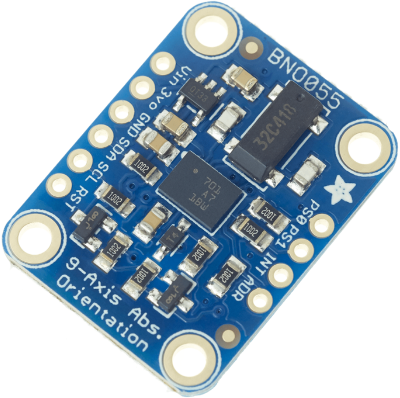
Need for calibration





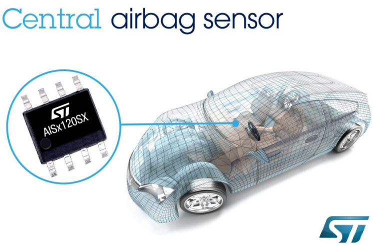
- Sensors inherently exhibit noise and biases.
- Raw data leads to unreliable outputs without correction.
- Proper filtering requires accurate mathematical models.
- Analysis includes working principles and noise sources.

Common terminology misconceptions



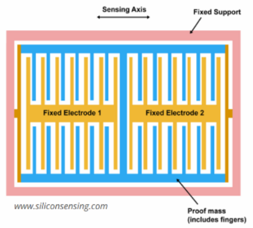
- "6-axis gyroscope" is technically an inaccurate term.
- "9-axis IMU" refers to 3x Accel, 3x Gyro, 3x Mag.
- IMUs do not have intrinsic "degrees of freedom" (DoF).
- They help estimate a body's 6 DoF state.

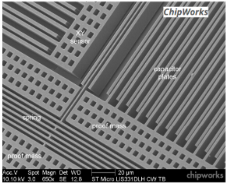
The rise of MEMS technology

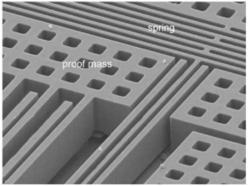


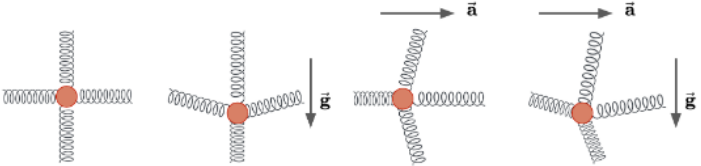
- Driven by automotive airbag mandate requirements.
- Accelerometers detect sudden vehicle deceleration.
- Mass production reduced costs significantly.
- Modern units feature real-time thermal calibration.

Principle: MEMS Accelerometers



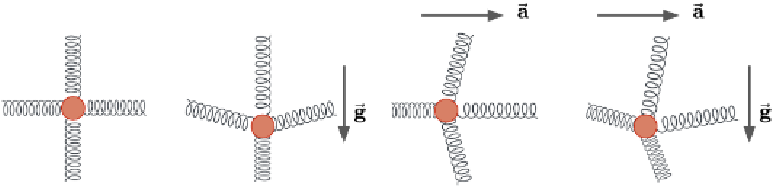






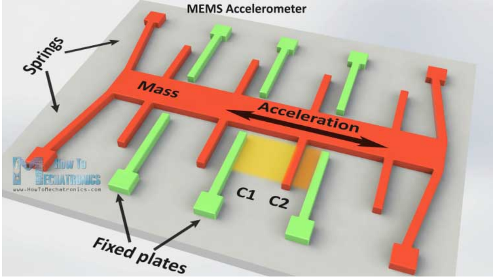
- Based on microfabricated mass-spring systems.
- A proof mass is suspended by springs in an envelope.
- External forces or gravity cause mass displacement.
- This results in measurable spring deformation.

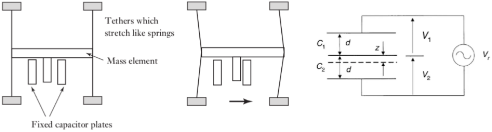
Gravity vs. linear acceleration



- Sensors *cannot* distinguish gravity from motion.
- Stationary sensors under gravity deflect like moving ones.
- Static deflection is used for roll/pitch estimation.
- Filtering is required to extract pure motion.

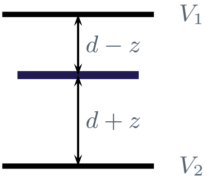
Capacitive Sensing Mechanism





- Uses interdigitated comb-like capacitor plates.
- Displacement alters the capacitance between plates.
- Relation: $\frac{\Delta C}{C} = -\frac{x/d}{1+(z/d)}$.
- Nonlinearities in z introduce sensing complexity.

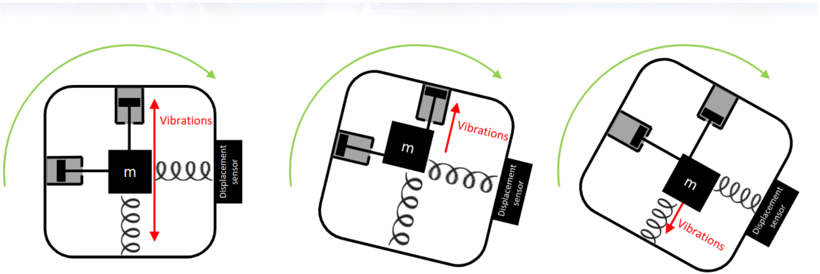
Linearizing with differential capacitance



- Voltage V_1, V_2 measured across gaps[cite: 111].
- Gap d changes by displacement z[cite: 111].
- Voltage: $V_1 - V_2 = \frac{C_2 - C_1}{C_1 + C_2}$.
- Resulting output: $V_1 - V_2 = V_r \frac{z}{d}$.
- Voltage is proportional to mass displacement.
- Newton-based model:

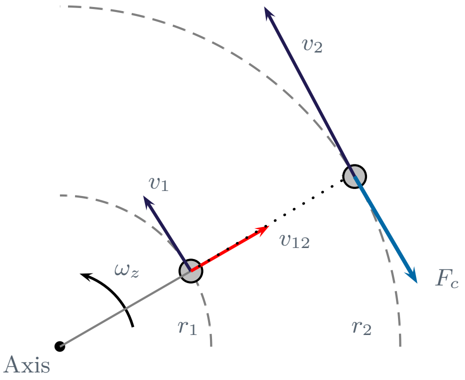
$$m\ddot{z}(t) + b\dot{z}(t) + kz(t) = F(t)$$

Vibrating gyroscope



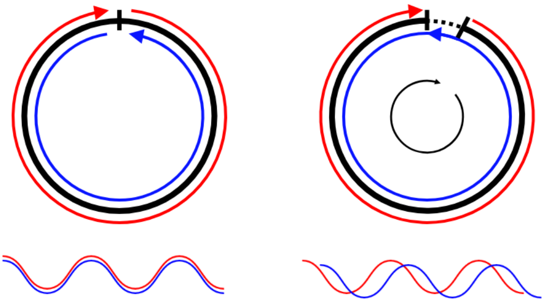
- An accelerometer placed away from axis of rotation in a rotating body can be used to measure angular acceleration.

Coriolis Force Model



- Centripetal acceleration: $a_c = \frac{v^2}{r}$
- Coriolis force: $F_c = -2m\omega_z \times v_{12} \quad \Rightarrow \quad \|\omega_z\| = \frac{\|F_c\|}{2m\|v_{12}\|}$.

Fiberoptic gyroscope



- At rest, the emitted frequencies (or wavelengths) are equal, since the cavity length is the same in both directions.
- When it is rotated, there is a small difference of cavity lengths because of the *Sagnac-Laue* effect, which yields a frequency difference between both output beams:

$$\Delta f_R = \frac{4A}{\lambda P}\omega_z$$

  where A and P are the area and perimeter of the ring cavity, λ is the light wavelength, and Ω_z is the angular speed.

Measurement model

Accelerometer Model:

$$K_a\tilde{a} = (I + S_a + M_a)a + b_a + T_a\Delta T + \nu_a + \epsilon_a$$

- K_a: Sensitivity scale factor.
- S_a: Scaling factor distortion.
- M_a: Misalignment factor
- b_a: Bias
- $T_a\Delta T$: Temperature-dependent variation.
- ν_a: White noise; ϵ_a: Random walk.

Gyroscope Model:

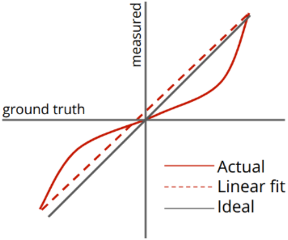
$$K_g\tilde{\omega} = (I + S_g + M_g)\omega + b_g + G_g a + T_g\Delta T + \nu_g + \epsilon_g$$

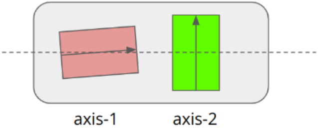
- K_g: Sensitivity scale factor.
- $G_g a$: Influence of linear acceleration.

Deterministic Error: Static Bias

- The most critical additive error in IMUs.
- Varies with temperature and power cycles.
- Results in a constant offset in measurements.
- Modeled as bias vector: $b_a = [b_{a,x}\ b_{a,y}\ b_{a,z}]^\top$.
- Often updated dynamically in Kalman filters.

Deterministic Errors: Distortion and misalignment





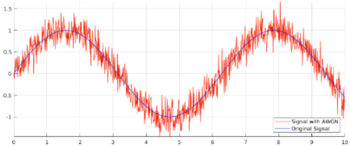
- Measurements may be distorted (scaled).
- Represented by a diagonal matrix $S_a = \text{diag}(S_{a,x}, S_{a,y}, S_{a,z})$.
- Corrects single-axis gain inaccuracies.
- Misalignment: axes are not perfectly orthogonal.
- Modeled with a 3x3 skew-symmetric matrix M_a.
- Accounts for cross-coupling between axes.
- Vital for multi-axis sensor data integrity.

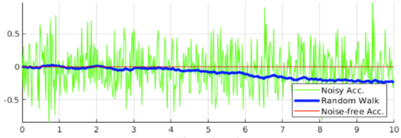
Gyroscope Specific: g-Sensitivity

- Gyros should ideally only measure angular rates.
- Linear acceleration can influence measurements.
- Modeled by a 3x3 matrix G_g.
- Error computed as $e_g = G_g a$.
- Critical in high-acceleration maneuvers.

Temperature-Dependent Bias

- MEMS structures are sensitive to heat.
- Material changes alter spring responses.
- Mitigated using internal Lookup Tables (LUTs).
- Monitoring ensures real-time correction.





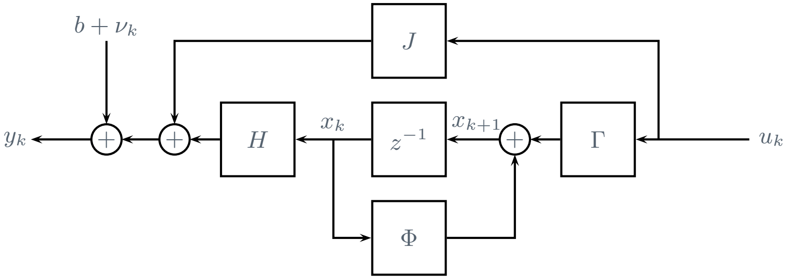
Random noise

- Arises from thermo-mechanical fluctuations.
- Modeled as additive White Gaussian noise.

Bias Random Walk

- Caused by electronic flicker noise.
- Leads to long-term drift in bias terms.
- Cannot be corrected via one-time calibration.
- Requires continuous EKF or UKF estimation.
- Angle Random Walk (ARW) for gyros, Velocity Random Walk (VRW) for accelerometers.

Simplified sensor models



- Used for stable thermal or low-dynamic environments; ignores negligible misalignment/g-sensitivity.
- Accelerometer: $y = \tilde{a} = a + b_a + \nu_a$.
- Gyroscope: $y = \tilde{\omega} = \omega + b_g + \nu_g$.

Kalman filter with bias estimate
Accelerometer

- Accelerometer: $y = \tilde{a} = a + b_a + \nu_a$.
- Model acceleration as low-pass filtered white noise:

$$a_{k+1} = \phi a_k + w_{a,k}, \quad w_a \in \mathcal{N}(0, \sigma_{w_a}^2)$$

- Dynamic-kinematic model based on Forward Euler discretization:

$$\begin{bmatrix} x_{k+1} \\ v_{k+1} \end{bmatrix} = \begin{bmatrix} 1 & T_s \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_k \\ v_k \end{bmatrix} + \begin{bmatrix} 0 \\ T_s \end{bmatrix} a_k$$

- Model the bias as a slowly varying white noise process:

$$b_{k+1} = b_k + w_{b,k}, \quad w_b \in \mathcal{N}(0, \sigma_{w_b}^2)$$

- All in all:

$$\begin{bmatrix} x_{k+1} \\ v_{k+1} \\ a_{k+1} \\ b_{k+1} \end{bmatrix} = \begin{bmatrix} 1 & T_s & 0 & 0 \\ 0 & 1 & T_s & 0 \\ 0 & 0 & \phi & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_k \\ v_k \\ a_k \\ b_k \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \sigma_{w_a} & 0 \\ 0 & 0 & 0 & \sigma_{w_b} \end{bmatrix} \bar{w}_k$$

$$y_k = \begin{bmatrix} 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_k \\ v_k \\ a_k \\ b_k \end{bmatrix} + \sigma_\nu \bar{\nu}_k$$

Model:

$$\chi_{k+1} = \Phi\chi_k + Q^{1/2}\bar{w}_k, \quad y_k = H\chi_k + R^{1/2}\bar{\nu}_k$$

Kalman filter:

- Initialization: $P_{0|-1} = I, \hat{\chi}_{0|-1} = 0.$
- Measurement:

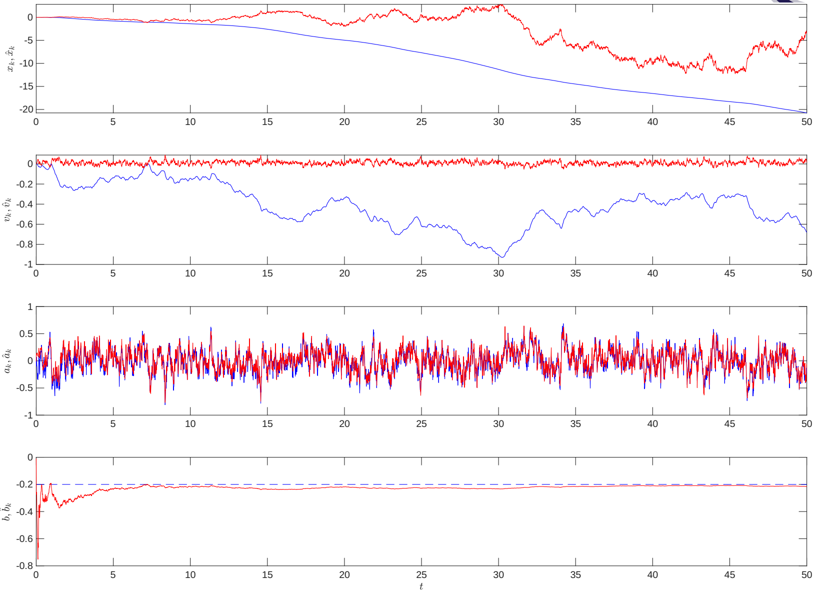
$$\begin{aligned}
\tilde{y}_{k|k-1} &= y_k - H\hat{\chi}_{k|k-1} \\
K_k &= P_{k|k-1}H^\top(HP_{k|k-1}H^\top + R)^{-1} \\
\hat{\chi}_{k|k} &= \hat{\chi}_{k|k-1} + K_k\tilde{y}_{k|k-1} \\
P_{k|k} &= (I - K_kH)P_{k|k-1}(I - K_kH)^\top + K_kRK_k^\top
\end{aligned}$$

- Prediction:

$$\begin{aligned}
\hat{\chi}_{k+1|k} &= \Phi\hat{\chi}_{k|k} \\
P_{k+1|k} &= \Phi P_{k|k}\Phi^\top + Q
\end{aligned}$$

Kalman filter with bias estimate

1D accelerometer, perfect model/estimator match



System dynamics:

$$\chi_{k+1} = \Phi_{3\times 3}\chi_k + \Gamma a_k, \quad y_k = H_{1\times 3}\chi_k + b + \sigma_\nu \bar{\nu}_k$$

where

$$\Phi_{3\times 3} = \begin{bmatrix} 1 & T_s & 0 \\ 0 & 1 & T_s \\ 0 & 0 & 0 \end{bmatrix}, \quad \Gamma = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad H_{1\times 3} = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

Kalman filter is unchanged! (still 4 states):

- Initialization: $P_{0|-1} = I, \hat{\chi}_{0|-1} = 0$.
- Measurement:

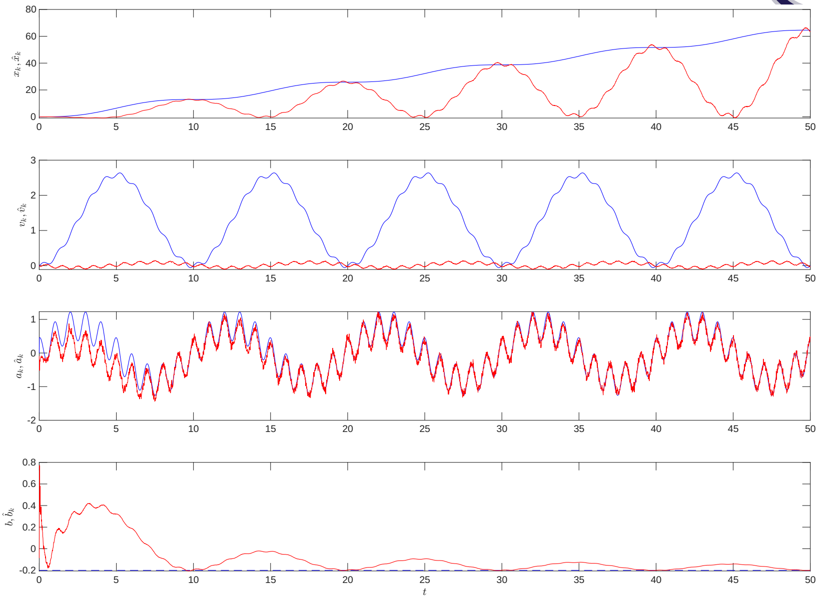
$$\begin{aligned} \tilde{y}_{k|k-1} &= y_k - H\hat{\chi}_{k|k-1} \\ K_k &= P_{k|k-1}H^\top (HP_{k|k-1}H^\top + R)^{-1} \\ \hat{\chi}_{k|k} &= \hat{\chi}_{k|k-1} + K_k\tilde{y}_{k|k-1} \\ P_{k|k} &= (I - K_kH)P_{k|k-1}(I - K_kH)^\top + K_kRK_k^\top \end{aligned}$$

- Prediction:

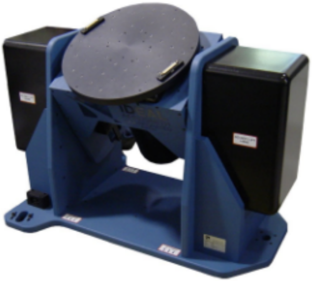
$$\begin{aligned} \hat{\chi}_{k+1|k} &= \Phi\hat{\chi}_{k|k} \\ P_{k+1|k} &= \Phi P_{k|k}\Phi^\top + Q \end{aligned}$$

Kalman filter with bias estimate

1D accelerometer, externally driven acceleration



Calibration: Professional Setup



- Requires precise and repeatable environments.
- Tumble tests use turntables for face data.
- Measurements correspond to $\pm 1g$.
- Procedures standardized by IEEE (e.g., IEEE 2019).

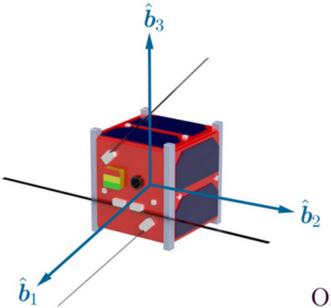
Calibrated Measurements

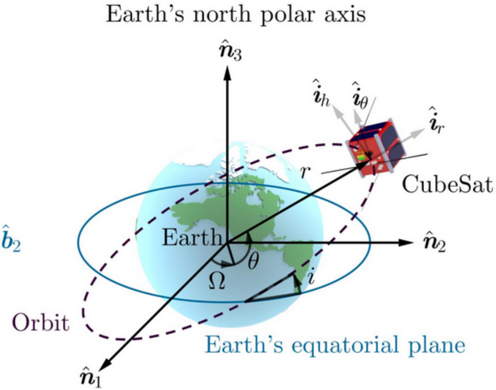
After calibration, the 'true' measurements may be produced by inverting the sensor model equations:

- $a = (I + S_a + M_a)^{-1}(K_a \tilde{a} - b_a - T_a \Delta T)$.
- $\omega = (I + S_g + M_g)^{-1}(K_g \tilde{\omega} - b_g - G_g a - T_g \Delta T)$.

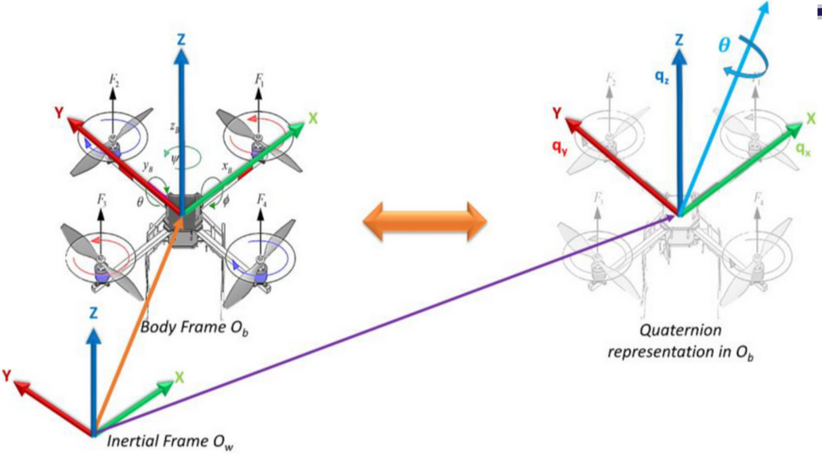
Simplified Calibration Equations:

- Simplified Gyro: $\omega = \tilde{\omega} - b_g$.
- Simplified Accelerometer: $a = \tilde{a} - b_a$.
- Note that stochastic noise *must* be filtered away first.





- Historical approach for satellite control.
- Requires complex trigonometric calculations.
- Impractical for early low-power processors.
- Suffers from Gimbal Lock singularities.



- Compact and singularity-free representation.
- Ideal for spacecraft and robotics.
- Facilitates fusion via averaging.
- Standard for efficient modern processing.

Complementary filters:

- *Fuse* high-frequency gyro data with low-frequency accelerometer data.
- Nonlinear versions provide robust attitude.
- Formulated on the orthogonal group $SO(3)$.
- Adaptive handling of gyro bias drift.

Madgwick filters:

- Efficient quaternion-based alternative to Kalman filters.
- No requirement for complex matrix inversion.
- Reduced latency for real-time applications.
- Lower computational complexity than EKF.

Summary

- Strengths of IMUs:
  - High sample rates (up to 1000 Hz).
  - Good for short-term state estimation.
  - Low cost for consumer MEMS units (2-5 EUR).
  - Indispensable for drone and robot navigation.
- Limitations of IMUs
  - Inherent noise and bias lead to drift.
  - Temperature and motion induce complex errors.
  - Long-term estimation requires external sensors.
  - Performance depends on calibration quality.
- Modeling deterministic/stochastic errors is key.
- Fusion frameworks compensate for weaknesses.
- MEMS technology makes IMUs cheap and ubiquitous.